

Training EquiUNet vs UNet on real CCP data

Methodology and results for the Shape2Fate detection backbone

Petr Vojtášek

June 3, 2026

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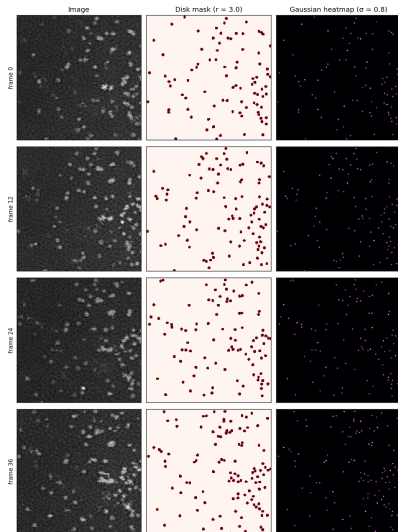
Question

For the Shape2Fate CCP detection backbone, does rotational equivariance pay off when training directly on real TIRF-SIM data?

Stack	Role	Frames	Annotators	Pts/ann. (in ROI)
033 (RPE-1)	train/val (80/20 contiguous)	120	3	13.5 k / 11.6 k / 13.3 k
012 (other cell)	held-out test	722	1	35.7 k (438 tracks)

- ▶ ROI: $[y=512:768, x=256:512]$, 256×256 px
- ▶ Per frame: ~ 110 – 120 CCP per annotator
- ▶ Hyperparameters (lr, wd) inherited from synthetic-data hparam sweep

Mask examples — what the network sees



Columns

- ▶ raw SIM-reconstructed image (ROI crop)
- ▶ disk mask ($r=3$, alternative target)
- ▶ fused Gaussian mask ($\sigma=0.8$, training target)

Rows — 4 representative frames from the 033 val stack.

Mask rendering

$$M(x, y) = \min\left(\sum_{c \in \mathcal{C}} e^{-\frac{(x-c_x)^2 + (y-c_y)^2}{2\sigma^2}}, 1\right)$$

Each fused centre $c \in \mathcal{C}$ contributes a Gaussian; sum clamped to keep the BCE target in $[0, 1]$.

Multi-annotator fusion — two algorithms compared

Greedy (operational default)

Require: pooled points $p_1, \dots, p_N$ in shuffled order;

threshold τ

```
1:  $\mathcal{K} \leftarrow \emptyset$ 
2: for each  $p$  do
3:   if  $\mathcal{K} = \emptyset$  then
4:     start new cluster  $\{p\}$ 
5:   else
6:      $j \leftarrow \arg \min_k \|\bar{c}_k - p\|_2$ 
7:     if  $\|\bar{c}_j - p\|_2 < \tau$  then
8:       append  $p$  to cluster  $j$ , update  $\bar{c}_j$ 
9:     else
10:      start new cluster  $\{p\}$ 
11:    end if
12:  end if
13: end for
14: return  $\{\bar{c}_k\}$ 
```

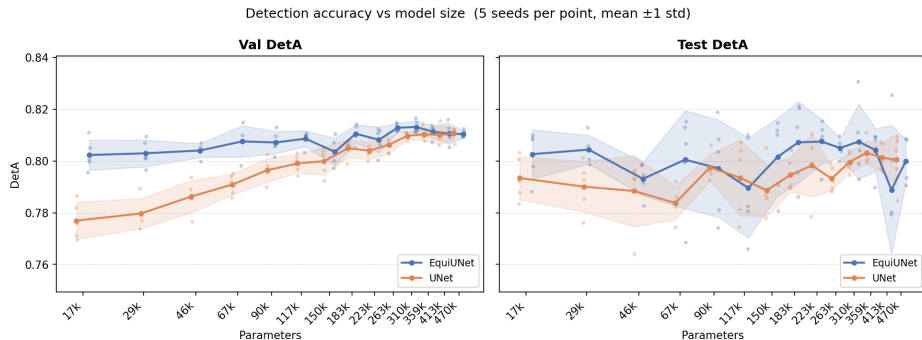
Agglomerative complete-linkage

Require: pooled points P ; threshold τ

```
1: clusters  $\leftarrow \{\{p\} : p \in P\}$  {singletons}
2: loop
3:    $(A, B) \leftarrow \arg \min_{X, Y} d_{\max}(X, Y)$ 
4:   where  $d_{\max}(X, Y) = \max_{a \in X, b \in Y} \|a - b\|_2$ 
5:   if  $d_{\max}(A, B) < \tau$  then
6:     merge  $A$  and  $B$ 
7:   else
8:     stop
9:   end if
10: end loop
11: return cluster centroids
```

	order-dependent?	complexity	cluster diameter
Greedy	yes (shuffled per-frame, seed = frame)	$\mathcal{O}(NK)$	may exceed τ (chain merge)
Complete-linkage	no	$\mathcal{O}(N^2 \log N)$	$\leq \tau$ by construction

Equivariance motivation — synthetic capacity ablation



5 seeds per point, mean ± 1 std. Val/test on real CCP frames.

- ▶ EquiUNet leads by $\sim +0.03$ DetA at ~ 17 k params $\Rightarrow$ equivariance as a **parameter-efficient prior**
- ▶ Both saturate at ~ 0.81 from ~ 110 k params $\Rightarrow$ **data/label-noise ceiling, not capacity**
- ▶ Does the benefit hold when *training directly on real data*?

Training pipeline

	UNet	EquiUNet
start filters / depth	16 / 3	16 / 3
parameters	≈ 450 k	≈ 489 k
symmetry group	—	SO(2), fourierbn act.
conv kernel	3×3	5×5
learning rate	$1.6 \cdot 10^{-4}$	$4.7 \cdot 10^{-3}$
weight decay	0.05	0.082
shared schedule: AdamW, BCE loss, 100 epochs, batch 16, 128×128 patches, val every epoch, 5 seeds		

- Inference: full 1024×1024 frame in one forward pass (fully convolutional), peak-finding, ROI filter post-hoc

Main result — UNet vs EquiUNet on merged real labels

Model	Val DetA	Test DetA
UNet sf=16	0.7783 ± 0.0118	0.8071 ± 0.0218
EquiUNet sf=16	0.7831 ± 0.0059	0.8116 ± 0.0150
Δ (Equi – UNet)	+0.0048	+0.0045
Welch p	0.45 (<i>n.s.</i>)	0.72 (<i>n.s.</i>)

- ▶ Point estimate equi leads, but **within seed noise at $n = 5$**
- ▶ **Seed std $\sim 2\times$ smaller for EquiUNet** $\Rightarrow$ symmetry stabilises optimisation
- ▶ Equivariance becomes statistically significant under noisier-mask conditions (fusion ablation, next slide)

Fusion-algorithm ablation — architecture robustness

5 seeds per cell, val DetA.

	Greedy $\tau=5$	Complete $\tau=5$	Complete $\tau=4$
UNet	0.7783 ± 0.0118	0.7620 ± 0.0127	0.7463 ± 0.0069
EquiUNet	0.7831 ± 0.0059	0.7855 ± 0.0040	0.7685 ± 0.0065
Welch p (Equi – UNet)	0.45 (<i>n.s.</i>)	0.012 *	< 0.001 ***

Within-architecture comparison	Welch p
UNet: greedy $\rightarrow$ complete-linkage (both $\tau=5$)	0.069 (marginal)
EquiUNet: greedy $\rightarrow$ complete-linkage (both $\tau=5$)	0.48 (<i>n.s.</i>)
UNet: $\tau=5 \rightarrow \tau=4$ (complete-linkage)	0.051 (marginal)
EquiUNet: $\tau=5 \rightarrow \tau=4$ (complete-linkage)	0.002 **

Equivariance prior absorbs the algorithm choice; the threshold remains a universal lever.

Cross-annotator generalisation — per-direction breakdown

Same training pipeline, but mask = single annotator (no fusion), val DetA scored per held-out annotator. 5 seeds $\times$ 3 train annotators = 15 runs / architecture.

train	eval	IAA (humans)	UNet	Δ vs IAA	EquiUNet	Δ vs IAA
ann1	ann2	0.740	0.693 ± 0.016	-0.047	0.718 ± 0.009	-0.022
ann1	ann3	0.748	0.752 ± 0.013	+0.003	0.770 ± 0.011	+0.022
	<i>ann1</i> $\rightarrow$	0.744	0.723	-0.022	0.744	0.000
ann2	ann1	0.740	0.842 ± 0.014	+0.102	0.851 ± 0.010	+0.111
ann2	ann3	0.730	0.777 ± 0.023	+0.047	0.782 ± 0.004	+0.052
	<i>ann2</i> $\rightarrow$	0.735	0.810	+0.075	0.817	+0.082
ann3	ann1	0.748	0.845 ± 0.007	+0.096	0.839 ± 0.016	+0.091
ann3	ann2	0.730	0.705 ± 0.015	-0.025	0.721 ± 0.018	-0.009
	<i>ann3</i> $\rightarrow$	0.739	0.775	+0.036	0.780	+0.041
<i>global aggregate</i>		0.7393	0.7688 ± 0.0394	+0.0295	0.7802 ± 0.0316	+0.0409

- **Networks exceed the human inter-annotator ceiling** in 4/6 directions $\Rightarrow$ network acts as a *label denoiser*
- Predicting ann2 (strict outlier, 11.6 k pts) is the only direction where networks underperform IAA; EquiUNet shrinks the gap

Out-of-domain robustness — transfer to DYNAMIN/RPE

Test stack disjoint from training (RPE1-egfpCLCa 033 → DYNAMIN/RPE 195), 151 frames, full HOTA pipeline. 5 seeds per config.

Config	HOTA	DetA
Synth UNet	0.777 ± 0.013	0.856 ± 0.011
Synth EquiUNet	0.760 ± 0.019	0.839 ± 0.018
Real EquiUNet	0.650 ± 0.025	0.700 ± 0.029
Real UNet	0.617 ± 0.047	0.644 ± 0.055

Welch t -test (DetA)	Δ	p
Synth UNet vs Real UNet	+0.212	0.0008 ***
Synth EquiUNet vs Real EquiUNet	+0.139	<0.0001 ***
Synth UNet vs Synth EquiUNet	+0.017	0.11 (<i>n.s.</i>)

Synth UNet wins. Synthetic training generalises to OOD data; equivariance prior brings no benefit once the training distribution is diverse.

Summary

1. Real-data training pipeline for the Shape2Fate detection backbone: 3-annotator greedy fusion ($\tau=5$, deterministic shuffle), Gaussian mask $\sigma=0.8$, BCE loss.
2. On merged labels (5 seeds, greedy $\tau=5$): EquiUNet point estimate leads UNet, but **+0.005 val DetA is within seed noise (Welch $p = 0.45$)**; seed std $\sim 2\times$ smaller for EquiUNet.
3. Fusion-algorithm ablation: **equivariance is statistically robust to upstream label noise**; complete-linkage gives EquiUNet +0.024 over UNet (Welch $p \leq 0.012$).
4. Cross-annotator: networks **exceed human inter-annotator agreement** (0.78 vs 0.74 IAA) — network as label denoiser.
5. **Out-of-domain robustness (DYNAMIN/RPE)**: synthetic-trained models beat real-trained by +0.14 to +0.21 DetA ($p \leq 0.001$); architecture irrelevant within synth ($p = 0.11$).